

Real_LTSpice_Netlist_1

Prompt :

User:

i'm trying to build an LTSpice AM Radio and here is the component list :

Listes Composantes :

Antenne: antenna wire (if no LC tuning then need to add it)

RF:

1× NPN transistor (BC547 / 2N2222 / 2N3904)

1× 4.7 kΩ resistor

1× 1 kΩ resistor

1× 100 kΩ resistor

2× capacitors (10–100 nF)

1× 9V battery

1× breadboard + wires

Detector:

Diode (1N4148 cheap for beginners project or IN34A)

Resistor (10 kΩ – 100 kΩ)

Capacitor (100 nF – 470 nF (0.1 μF – 0.47 μF))

ou $1 / RF \ll RC \ll 1$ / Audio (RF= rf period, Audio= audio output)

Audio Amplifier:

Transistor Q1 (2N3904, BC547, BC337)

R1 (base bias): 10 kΩ

R2 (collector load): 4.7 kΩ

Re (emitter resistor): 1 kΩ

C1 Input coupling capacitor: 10 μF (electrolytic or film)

C2 Output coupling capacitor: 10 μF (electrolytic or film)

Depending on your intended output:

->Headphones (recommended for beginners):

Impedance: ~32 Ω

Simple earbuds or small headphones work fine.

->Small Speaker (optional-Requires more current → may need a stronger amplifier stage later)

Size: 3–5 inches

Impedance: 8–16 Ω

f= 198kHz (BBC radio 4)

f = 693kHz (BBC 5 live)

f = 1467kHz (Trans world radio)

i want you to make a netlist for it , and also make it possible to simulate it(generate a sinusoidal wave before the circuit to simulate it).

Answer

Assistant:

This is a classic, elegant AM receiver design. To make it work reliably in LTSpice, I had to make a few specific engineering decisions based on your component list:

- The AM Signal Generator:** Standard SPICE voltage sources don't have a simple "AM Radio" button. I used an arbitrary behavioral voltage source (`B1`) to mathematically multiply a 693 kHz carrier wave (BBC 5 Live) with a 1 kHz audio test tone.
- Collector-Feedback Biasing:** Your list has one base resistor per stage (100 kΩ for RF, 10 kΩ for Audio). If connected straight to the 9V battery, the transistors would instantly saturate and distort. By connecting these resistors from the *collector* to the base, the transistors self-bias beautifully into the active amplification region.
- Pre-Biased Detector:** The diode is connected directly to the RF amplifier's collector. The transistor's DC voltage naturally pushes the diode right to the edge of its 0.6V threshold. This makes the diode incredibly sensitive to tiny microvolt antenna signals!

The RC Time Constant Math

You correctly noted the critical envelope detector rule: $1/f_{RF} \ll RC \ll 1/f_{Audio}$

Let's plug in the numbers for a 693 kHz carrier and a 1 kHz audio signal:

- RF Period ($1/f_{RF}$):** $1 / 693,000 = 1.4 \text{ }\mu\text{s}$
- Audio Period ($1/f_{Audio}$):** $1 / 1,000 = 1,000 \text{ }\mu\text{s}$

If we use your $10 \text{ k}\Omega$ resistor and a 10 nF capacitor, the RC time constant is $100 \text{ }\mu\text{s}$.

$1.4 \text{ }\mu\text{s} \ll 100 \text{ }\mu\text{s} \ll 1000 \text{ }\mu\text{s}$

(Note: Using a 100 nF capacitor gives $RC = 1000 \text{ }\mu\text{s}$, which equals the audio period and would severely muffle our 1 kHz test tone. I've set it to 10 nF in the simulation for the clearest audio waveform).

The LTSpice Netlist

```
* AM Radio Receiver Simulation (693 kHz BBC 5 Live)
* --- Power Supply ---
Vcc VCC 0 9V
```

```

* --- RF Antenna / Signal Source ---
* B1 creates the AM modulated signal: 693kHz carrier, 1kHz audio tone, 10mV
amplitude
B1 IN 0 V=(1+0.8*sin(2*pi*1k*time))*sin(2*pi*693k*time)*10m
R_ant IN RF_IN 50

* --- LC Tuning Tank (Added for simulation) ---
* Tunes to approx 693 kHz
L1 RF_IN 0 220u
C_tune RF_IN 0 240p

* --- RF Amplifier Stage (Common Emitter) ---
C_rf_in RF_IN RF_B 10n
R_rf_b RF_C RF_B 100k ; Collector-feedback bias
R_rf_c VCC RF_C 4.7k
R_rf_e RF_E 0 1k
C_rf_e RF_E 0 100n ; Emitter bypass for high AC gain
Q1 RF_C RF_B RF_E 2N3904

* --- AM Detector Stage ---
* Directly coupled to utilize the RF stage's DC offset for pre-biasing
D1 RF_C DET_OUT 1N4148
R_det DET_OUT 0 10k
C_det DET_OUT 0 10n ; 10nF perfectly balances the RC formula

* --- Audio Amplifier Stage ---
C1 DET_OUT AF_B 10u
R_af_b AF_C AF_B 10k ; Collector-feedback bias
R_af_c VCC AF_C 4.7k
R_af_e AF_E2 0 1k
Q2 AF_C AF_B AF_E2 2N3904
C2 AF_C OUT 10u
R_load OUT 0 32 ; 32-ohm Headphones

* --- Transistor & Diode Models ---
.model 2N3904 NPN(IS=1E-14 VAF=100 BF=300 IKF=0.4 XTB=1.5 BR=4 CJC=4p CJE=8p
RB=20 RC=0.1 RE=0.1 TR=250p TF=350p ITF=1 VTF=2 XTF=3 VCE0=40)
.model 1N4148 D(Is=2.52n Rs=.568 N=1.752 Cjo=4p M=.4 tt=20n Iave=200m Vpk=75)

* --- Simulation Setup ---
.tran 0 5m 0 100n
.end

```

How to Run This in LTSpice

1. **Create the file:** Open Notepad (or any plain text editor), paste the entire code block above, and save the file as `AM_Radio.cir` somewhere easily accessible.
2. **Open in LTSpice:** Launch LTSpice, go to **File -> Open**, change the file type dropdown to "All Files (.)", and open your `AM_Radio.cir` file.
3. **Run the Simulation:** Click the **Run** icon (the running figure). LTSpice will process the netlist and open a blank black waveform viewer.
4. **Probe the signals:** Right-click the black screen, select **Add Traces**, and select `V(in)` to see your raw AM antenna signal. Then add `V(det_out)` to see the extracted envelope, and finally `V(out)` to see the amplified 1 kHz audio ready for the headphones!

[Real_LTSpice_Netlist_2](#)